

Fig. 1: Semi-active radar homing guidance basic concept

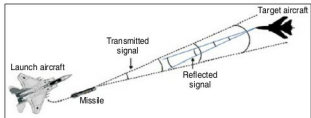


Fig. 2: Active radar guidance basic concept

radar. A radar receiver onboard the missile receives the radar energy reflected off the target and locks on to the target.

Once the missile is locked on to target, it may be launched. The radar seeker unit provides the angular error information to the flight path control system, which steers the missile to intercept the target through tail fins.

Semi-active radar homing guidance uses a continuous wave doppler radar in bi-static configuration. It uses bi-static because transmit and receive antennae are at different locations with the former being collocated with a fire control radar either on the launch aircraft or on ground, and the latter on the nose cone of the missile.

Doppler shifted receive signal frequency is computed from the closing velocity determined using flight path geometry. Information on the receive signal frequency is used by the missile to acquire the target. Fig. 1 shows semi-active radar homing basic concept when both fire control radar and missile are onboard the launch aircraft.

Semi-active radar guided weapons have the advantages of being low-cost and less complex. Due to the fact that the fire control radar needs to remain committed throughout the duration of the engagement till the

target intercept has been achieved, semi-active radar guidance has certain disadvantages, more so when the radar is located on the launch aircraft. It renders the launch aircraft vulnerable to counter attack from the adversary using anti-radiation missiles as the radar itself acts as a beacon for the counter attacking missile.

Despite this shortcoming, semi-active radar guidance is likely to stay because of better radar resolution and higher power possible with ground based or aircraft-mounted radars.

It may be mentioned here that radar resolution becomes better with an increase in transmit antenna size, and the radar's size and weight have a direct bearing on power output capability of the radar. A large-size transmit antenna and large-size radar cannot be accommodated within the missile as would be the requirement in active radar guidance.

Raytheon's AIM-7 Sparrow with an operational range of 30km to 50km for different variants uses semi-active radar homing guidance.

Active radar guidance

In the case of active radar guidance, the missile tracks its target by means of emissions that it generates itself with the radar transceiver located onboard the missile. The missile has the required electronics hardware and software to find and track the target autonomously and does not depend upon the services of an external radar transmitter as is the case in semi-active radar guidance.

Active radar guidance is commonly used for terminal homing in anti-ship, surface-to-air and air-to-air missiles. Fig. 2 shows the basic con-

cept of active radar homing guidance.

One advantage of active radar homing guidance is higher tracking accuracy and better kill probability as compared to semi-active radar guidance. This is due to the missile being much closer to the intended target in the terminal phase than would be the launch platform. This also gives the missile higher resistance to electronic countermeasures.

Another major advantage is the much-reduced vulnerability of the launch platform to counter attack. Since the missile is autonomous in the terminal phase, launch platform, ground based or on aircraft need not keep its radar enabled during this phase. In fact, the aircraft in case of being the launch platform can exit the scene or undertake other tasks while the missile homes on to the target providing fire-and-forget capability.

Looking at disadvantages, active radar guided missiles are likely to have increased size and weight as compared to missiles employing semi-active radar guidance. In the case of former, the missile has to accommodate the radar transceiver and electronics for autonomous target tracking.

Also, these missiles are more expensive as sophisticated electronics have only one-time use.

Another shortcoming is that, active radar guided missiles, due to radar emission, are more than likely to be detected by sophisticated radar-warning sensors equipping modern fighter aircraft, enabling these to take an evasive action or deploy countermeasures well in time. Given the fact that even modern aircrafts cannot match the manoeuvrability and agility of state-of-the-art missiles, there is not much an aircraft can do to avoid interception.

Due to restriction on size of the radar transmitter that is linked to power available at radar output and size of transmitting antenna that is linked to beam width, active radar guidance alone cannot provide guid-